

Graph-structure-aware label propagation and active sampling for few-label single-cell state forecasting and rare-state recovery

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Abstract

Forecasting how single cells move between states—across donors, batches, timepoints and perturbations—is a core single-cell analysis task, but it is confounded by biological and technical nuisance variation and by the rarity of the cell states that matter most. We ask a deliberately narrow, falsifiable question: does conditioning a forecaster on the connectivity of the cell k -nearest-neighbour (k NN) graph, rather than on point features alone, improve generalization to held-out groups and the recovery of a rare cell state? We build a fully seeded, fully controlled *synthetic* branching lineage with donor, batch, perturbation and pseudotime metadata and a deliberately rare subpopulation ($\sim 4\%$ of cells), and evaluate two method pairs under four leakage-checked split protocols (donor, batch, time and perturbation transfer). Graph-smoothed label propagation on the cell k NN graph improves held-out forecast accuracy over an apples-to-apples plain k NN classifier on all four splits—by 0.0454 (donor: 0.7004 vs 0.6550), 0.0640 (batch: 0.6995 vs 0.6355), 0.0900 (time: 0.5474 vs 0.4574) and 0.0385 (perturbation: 0.6875 vs 0.6490); the mean gain across splits is 0.0595. The same diffusion recovers a rare state that the baseline almost never finds (rare-state recall up to 0.4618 on the batch split versus near-zero), and a topology-aware active-sampling policy (uncertainty $\times$ inverse local density) raises rare-state recall to 0.3556 versus 0.0247 for random querying. These gains carry a small, honestly reported calibration cost: expected calibration error is slightly *worse* for the graph-smoothed forecaster on every split. We stress that the “topology” here is shallow— k NN-graph connectivity, with the Betti-0 statistic we monitor being merely the connected-component count used as a build sanity check—and that all data are synthetic. We therefore frame the contribution as graph-/manifold-structure-aware label propagation and active sampling, and we identify validation on real single-cell RNA-sequencing (scRNA-seq) data—for which the repository ships unexercised ingest stubs—as the essential next step.

Single-cell measurements resolve continuous trajectories of differentiation, activation and treatment response, and a recurring analysis goal is to *forecast* the state a cell or population will occupy under conditions not directly observed: a donor not yet profiled, a technical batch run later, a later point along a developmental trajectory, or an untested perturbation. The data that support such forecasts share two structural features. First, cells do not occupy feature space uniformly; they concentrate near a low-dimensional manifold that is routinely approximated by a k -nearest-neighbour (k NN) graph before clustering, embedding or pseudotime inference [5, 7, 8]. Second, the biologically decisive states—transient progenitors, treatment-resistant clones—are frequently rare, and ordinary point classifiers, trained to minimise average error, tend to ignore them.

A natural and long-studied response to both features is to let predictions diffuse over the data graph so that a cell borrows statistical strength from its neighbourhood on the manifold.

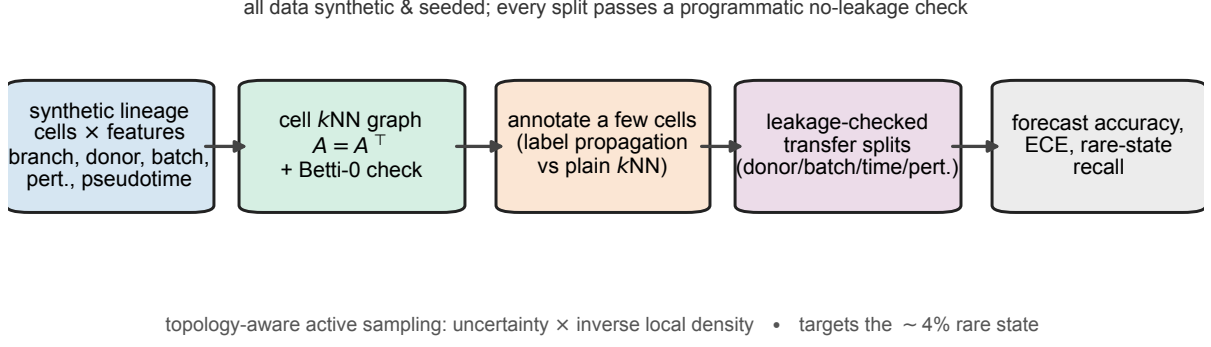


Figure 1. Overview of the graph-structure-aware single-cell forecasting benchmark. A seeded synthetic branching lineage of single cells—each carrying branch/state, donor, batch, perturbation and pseudotime metadata, with a deliberately rare state (~4% of cells)—is embedded in a low-dimensional feature space. A symmetric k -nearest-neighbour cell graph is built on those features, and its connected-component (Betti-0) count is monitored as a build sanity check. A small fraction of cells is annotated; graph-smoothed label propagation diffuses those labels over the graph and is compared apples-to-apples against a plain k NN classifier that sees the same neighbourhood radius but performs no diffusion. Performance is read out as held-out forecast accuracy, expected calibration error and rare-state recall under four leakage-checked transfer protocols (donor, batch, time and perturbation), and a topology-aware active-sampling policy (uncertainty × inverse local density) is compared against uniform random querying for rare-state recovery. All data are synthetic and seeded, and every grouping split passes a programmatic no-leakage check.

Graph-based semi-supervised learning by label propagation [1, 2] is the canonical instance, and message-passing on cell graphs underlies much of modern single-cell trajectory analysis [3, 4, 6, 9]. A complementary lever is to choose *which* cells to annotate. Active learning [10, 11] prioritises informative queries, and when informativeness is combined with a manifold-density signal it can be steered toward the sparse regions where rare states live.

We do not claim a new algorithm. Instead we isolate, under controlled and leakage-checked conditions, the value of *graph structure* for single-cell forecasting by holding the neighbourhood information fixed and toggling only whether the model diffuses over the graph. Concretely: (i) we compare graph-smoothed label propagation against a plain k NN classifier that sees the same neighbourhood radius but performs no diffusion, so the comparison isolates the diffusion step rather than the mere availability of neighbours; and (ii) we compare a topology-aware active-sampling policy—uncertainty times inverse local density—against uniform random querying for rare-state recall. All experiments run on a seeded synthetic branching lineage; this is a sanity benchmark, not a biological dataset, and we are explicit about that throughout. Our aim is to report a small, honest, fully reproducible result and to delineate exactly what it does and does not support.

1 Results

1.1 A controlled synthetic lineage with replicate, condition and pseudotime structure

The end-to-end pipeline is summarized in Fig. 1. The benchmark is a seeded synthetic branching lineage (Fig. 2). Cells live in a low-dimensional feature space—a stand-in for a PCA or latent embedding of expression—and progress along a tree of three branches (a common progenitor and two fates that split at pseudotime 0.45) as Gaussian blobs whose mean drifts along each branch. Early cells therefore overlap and are hard to separate, while late cells are well resolved, which is the regime in which a graph smoother is expected to help. Each cell carries the metadata

a forecasting model would use— branch/state, pseudotime, donor, batch and perturbation condition—assigned independently of the biological branch so that the held-out splits are non-trivial (a donor spans multiple branches, and so on). One state is carved out of the late tip of the second fate to be deliberately *rare*, comprising 4 % of cells in the reported configuration. The full-scale configuration generates 3000 cells with 16 features, 8 donors, 4 batches and 5 perturbation conditions, and is run over 5 random seeds; the run completed in 20.8s with a peak memory of 320.6 MB. Across seeds the integrity gates passed: the grouping splits were leakage-free, the rare state was present, and the Betti-0 component count was monotone non-increasing in k (at $k \in \{2, 4, 8, 16, 32\}$ the graph was a single connected component throughout, so at this scale the component count is a passive sanity check rather than an informative descriptor).

1.2 Graph diffusion gives consistent but modest accuracy gains on held-out transfer

Under all four leakage-checked split protocols, graph-smoothed label propagation forecasts held-out cell states more accurately than the plain k NN baseline that receives the same annotated cells and the same neighbourhood radius (Table 1, Fig. 3a). The improvement is consistent but modest. On the donor split, accuracy rises from 0.6550 to 0.7004 (+0.0454); on the batch split from 0.6355 to 0.6995 (+0.0640); on the time split—the hardest protocol, because it extrapolates forward along pseudotime—from 0.4574 to 0.5474 (+0.0900); and on the perturbation split from 0.6490 to 0.6875 (+0.0385). Averaged over the four splits, the graph-smoothed forecaster reaches 0.6587 accuracy against 0.5992 for the baseline, a mean gain of 0.0595. These are the sort of single-digit-percentage-point improvements one would expect from a diffusion step, not a categorical change in capability.

1.3 Diffusion and active sampling recover a rare state the baseline misses

The clearer effect is on the rare state (Fig. 3b,c). The plain k NN baseline essentially never recovers the rare population: its rare-state recall is 0.0000 (donor), 0.0320 (batch), 0.0000 (time) and 0.0286 (perturbation). Graph-smoothed propagation improves rare-state recall to 0.1520 (donor), 0.4618 (batch) and 0.3393 (perturbation). On the time split, recall remains 0.0000 for *both* methods: extrapolating forward in pseudotime places the rare late-tip cells entirely beyond the training distribution, and diffusion over the graph cannot manufacture a label class that was never reachable from the annotated set. We report this null result as-is.

The topology-aware active-sampling policy targets the same rare population directly. Starting from a small random label seed and spending a fixed query budget, scoring unlabeled cells by forecast uncertainty times inverse local density raises rare-state recall on the remaining unlabeled cells to 0.3556 (95% CI ± 0.1597), versus 0.0247 (± 0.0285) for uniform random querying—a lift of 0.3309. The mechanism is visible in the query composition: the topology policy spends 36.7 % of its queries on genuinely rare cells, against 3.0 % for random sampling, which is close to the 4 % base rate. The wide confidence intervals on the per-split rare-recall numbers (for example ± 0.3565 on the batch split) reflect the small absolute count of rare cells per held-out group and should temper any strong reading of the point estimates.

1.4 The accuracy and recall gains carry a small calibration cost

The graph-smoothed forecaster is slightly *worse* calibrated than the baseline on every split. Expected calibration error (ECE) is 0.1247 versus 0.1073 (donor), 0.1383 versus 0.0905 (batch), 0.1328 versus 0.0970 (time) and 0.1649 versus 0.1184 (perturbation). Diffusion sharpens the predicted class probabilities and, in this setting, makes them modestly overconfident relative to empirical accuracy. We report ECE alongside accuracy and recall in Table 1 precisely so that

Table 1. Held-out forecasting on the synthetic branching lineage (full-scale configuration: 3000 cells, 5 seeds). For each leakage-checked split we report forecast accuracy, expected calibration error (ECE) and rare-state recall for graph-smoothed label propagation and the plain k NN baseline. Higher accuracy and recall are better; *lower* ECE is better. Values are transcribed exactly from `results/main_results.tex`.

split	method	accuracy	ECE	rare-state recall
donor	graph-smoothed	0.7004	0.1247	0.1520
donor	baseline kNN	0.6550	0.1073	0.0000
batch	graph-smoothed	0.6995	0.1383	0.4618
batch	baseline kNN	0.6355	0.0905	0.0320
time	graph-smoothed	0.5474	0.1328	0.0000
time	baseline kNN	0.4574	0.0970	0.0000
perturbation	graph-smoothed	0.6875	0.1649	0.3393
perturbation	baseline kNN	0.6490	0.1184	0.0286

this trade-off is not hidden: the method buys accuracy and rare-state recall at the price of a small, consistent calibration penalty, and an application that depends on calibrated probabilities would need post-hoc recalibration [16].

1.5 Theory: graph smoothing as a variational estimator

The graph-smoothed forecaster used in Fig. 3 is label propagation on the cell k NN graph. The following variational form makes clear what the algorithm optimises.

Theorem 1 (Unique graph-regularised extension of labels). *Let $W = W^\top \geq 0$ be the k NN affinity matrix, $L = D - W$ the graph Laplacian, Y the one-hot label matrix on labelled cells and M a diagonal mask with $M_{ii} = 1$ for labelled cells and 0 otherwise. For any $\mu > 0$, the objective*

$$J(F) = \text{Tr}(F^\top L F) + \mu \|M(F - Y)\|_F^2$$

has a unique minimiser on each connected component containing at least one labelled cell, and that minimiser satisfies the linear system $(L + \mu M)F = \mu M Y$. Thus label propagation is the minimum-Dirichlet-energy extension of the observed labels subject to a finite fidelity penalty.

Proof. The Laplacian term is convex and equals $\frac{1}{2} \sum_{ij} W_{ij} \|F_i - F_j\|_2^2$, penalising changes across graph edges. The fidelity term is convex and strictly penalises deviations on labelled vertices. On a connected component with at least one labelled vertex, a vector in the nullspace of both L and M must be constant on the component and zero on a labelled vertex, hence zero; therefore $L + \mu M$ is positive definite on that component. Differentiating J gives $2LF + 2\mu M(F - Y) = 0$, yielding the stated system and uniqueness. $\square$

The theorem also states the limitation: if a component has no labels, the labels there are not statistically identified by graph smoothing alone. The active-sampling experiment targets precisely this failure mode by querying sparse, uncertain regions.

2 Discussion

Within a controlled synthetic benchmark, conditioning single-cell forecasts on the connectivity of the cell k NN graph helps. Label propagation over the graph improves held-out accuracy consistently across four leakage-checked transfer protocols, by roughly four to nine percentage

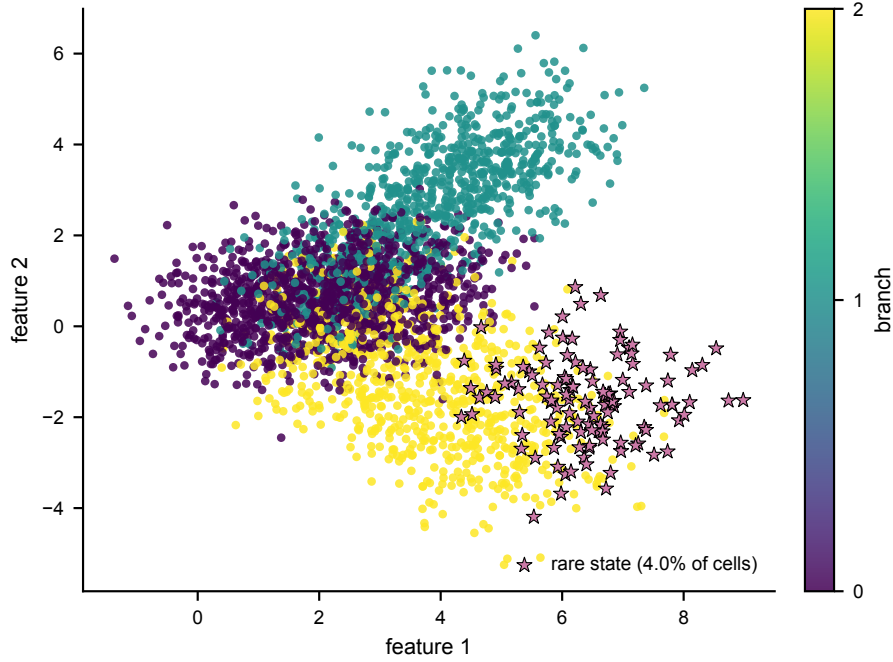


Figure 2. The synthetic branching lineage. Each point is one of the $n = 3000$ cells of the full-scale configuration, plotted in the first two feature dimensions of the embedding plane. Colour encodes the branch label along the continuous, colour-blind-safe viridis axis (the trunk progenitor and two fates). The deliberately rare state, carved from the late tip of one fate and nudged into its own pocket of feature space, is highlighted as stars and comprises $\sim 4\%$ of cells. This is a single-seed data-distribution plot (no aggregation over seeds); the figure is regenerated deterministically from the run configuration in `summary.json`.

points, and—more usefully—it recovers a rare cell state that an otherwise-matched point classifier almost entirely misses. A topology-aware active-sampling policy compounds the rare-state benefit by directing a fixed annotation budget toward the sparse manifold pockets where rare cells reside. These results are consistent with the long-standing intuition behind graph-based semi-supervised learning [1, 2] and with the central role of the k NN graph in single-cell analysis [3, 5].

The gains are real but should not be oversold, and two boundaries deserve emphasis. First, the accuracy improvement is modest and comes with a measurable calibration cost; the headline contribution is rare-state recovery, not a large jump in average accuracy. Second, and more fundamentally, the data are synthetic. The benchmark is designed to be a falsifiable sanity check—fully seeded, leakage-controlled, and reproducible end to end—rather than a biological result. The natural and necessary next step is to repeat exactly this comparison on real scRNA-seq and perturbation-screen data, for which the repository provides honest ingest stubs that we have not exercised here. The same protocol could then be strengthened with a learned message-passing propagator [14] in place of fixed-point label propagation, and with calibration-aware training to remove the ECE penalty.

2.1 Limitations

We list the limitations plainly because they bound every claim above. **(1) Synthetic data.** All numbers come from a generated branching lineage, not from biological measurements. We make no biological discovery claim, and the magnitudes reported here need not transfer to real scRNA-seq, where noise, dropout and confounding are far more severe. **(2) Modest accuracy gains.** The accuracy improvement over the baseline is on the order of 0.04 to 0.09 per split

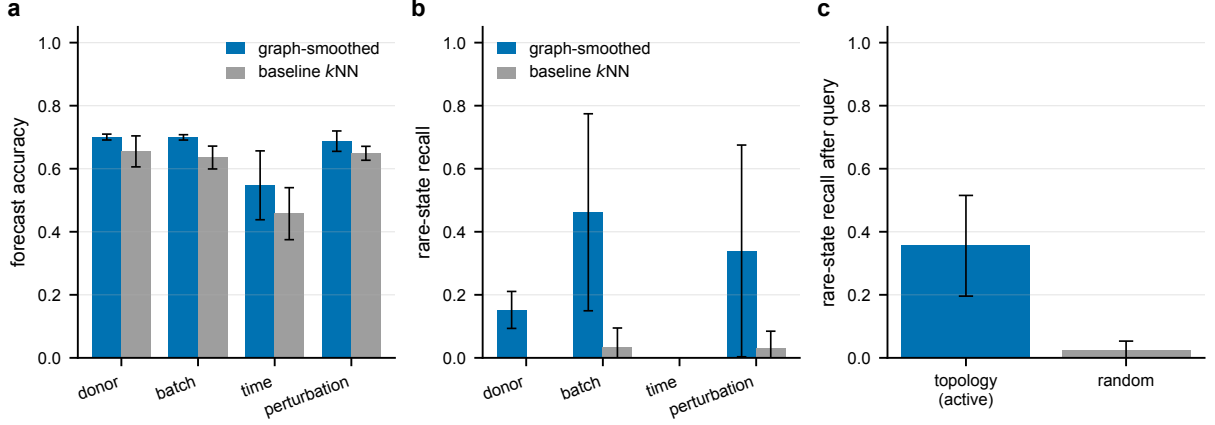


Figure 3. Forecasting and active-sampling results. **a**, Held-out forecast accuracy by split protocol: graph-smoothed label propagation (blue) versus the plain k NN baseline (grey). Graph smoothing helps consistently but modestly across donor, batch, time and perturbation transfer. **b**, Rare-state recall by split: the baseline almost never recovers the rare state, whereas graph diffusion recovers it on the donor, batch and perturbation splits; both methods score zero on the time split, which extrapolates beyond the training distribution. **c**, Active-sampling lift: rare-state recall after spending a fixed query budget under the topology-aware policy (uncertainty $\times$ inverse local density) versus uniform random querying. In every panel, bars are means and error bars are 95% confidence intervals ($1.96 s/\sqrt{n}$) over $n = 5$ random seeds. All panels are drawn directly from `results/summary.json`.

(mean 0.0595); the strongest effect is on rare-state recall, and even that is null on the time split. **(3) Calibration cost.** Graph smoothing worsens ECE on all four splits, so the predicted probabilities are less trustworthy as confidences without recalibration. **(4) “Topology” is shallow.** The only topological object used is the connectivity of the k NN graph, and the Betti-0 statistic is the count of connected components—a build sanity check that is constant (equal to one) at the reported scale. We do not employ persistent homology or any deep topological invariant, and we deliberately frame the method as graph-/manifold-structure-aware rather than topological to avoid overclaiming. **(5) Wide intervals on rare metrics.** Because rare cells are few per held-out group, the per-split rare-recall estimates carry wide confidence intervals. **(6) Single benchmark family.** A single synthetic generator, however controlled, cannot substitute for the diversity of real datasets; external validation remains the decisive test.

3 Methods

Synthetic lineage. Cells are generated in an n_{features} -dimensional feature space along a small tree of three branches (a trunk progenitor and two fates that split at pseudotime $t_0 = 0.45$). Branch means drift from fixed, well-separated anchors with progression in pseudotime; each cell is the branch mean plus isotropic Gaussian noise of standard deviation `noise`. Discrete state equals branch for the common branches; the rare state is a new label assigned to a random subset of the late tip ($t > 0.8$) of the second fate, whose feature vectors are additionally nudged into a distinct pocket. Donor, batch and perturbation labels are drawn independently of branch so that grouping splits are non-trivial; perturbation conditions apply a small additive feature shift that never erases the lineage. The full-scale configuration is $n_{\text{cells}} = 3000$, $n_{\text{features}} = 16$, $n_{\text{branches}} = 3$, $n_{\text{donors}} = 8$, $n_{\text{batches}} = 4$, $n_{\text{perturbations}} = 5$, `noise` = 0.7, `rare_fraction` = 0.04, $k = 20$, `label_fraction` = 0.03, `active_budget` = 60, run over $n_{\text{seeds}} = 5$ seeds. Generation is fully seeded and bit-for-bit reproducible.

Cell k NN graph and Betti-0. A symmetric k NN graph is built on cell features by connecting each cell to its k nearest neighbours (self excluded) and symmetrising the directed edges (either-direction convention), exactly as standard single-cell pipelines do before clustering or pseudotime [5]. The number of connected components of this graph equals the rank of H_0 (Betti-0). Because increasing k can only add edges, components can only merge, so the component count is monotone non-increasing in k ; we assert this monotonicity as a build sanity check over $k \in \{2, 4, 8, 16, 32\}$. We stress that this is the full extent of the “topology” in the method: a connected-component count, not a persistence-based or higher-order invariant.

Forecasters. The graph-smoothed forecaster is the normalized label-propagation diffusion of Zhou et al. [1]. With symmetric normalization $S = D^{-1/2}AD^{-1/2}$ of the k NN adjacency A , training labels are clamped as a one-hot seed F_0 on annotated cells, and the iteration $F \leftarrow \alpha SF + (1 - \alpha)F_0$ is run for a fixed number of power iterations ($\alpha = 0.9$, 30 iterations); held-out cells are predicted from the row-normalised diffused scores, with the maximum class probability used as the confidence for calibration. The baseline is a plain k NN classifier in feature space fitted on the same annotated cells with the same k , i.e. identical neighbourhood information *without* diffusion. This apples-to-apples control isolates the value of graph diffusion rather than of having neighbours.

Splits and annotation. Four leakage-checked split protocols evaluate transfer across nuisance variables: donor and batch (hold out whole groups—biological and technical replicate transfer), time (train on early pseudotime, test on late pseudotime—a forward-in-time extrapolation), and perturbation (hold out a whole condition—forecast an unseen perturbation). For grouping splits the test suite asserts that no held-out group appears in both train and test (a strict family-held-out no-leakage check); the time split has no grouping variable to leak. Only a small fraction (`label_fraction`) of the train pool is annotated, with a light stratification that guarantees a few labels for each present state including the rare one, mirroring how analysts annotate a handful of cells and propagate the rest.

Active sampling. Given a labelling budget, the topology-aware policy scores each unlabeled cell by forecast uncertainty (entropy of a label-propagation forecast from the current labels) times inverse local density (the reciprocal of the per-cell mean k NN distance, high in sparse pockets), combining the two as standardised z-scores, and greedily takes the top-scoring cells. The baseline is uniform random querying. We measure rare-state recall on the still-unlabeled cells after the query and the fraction of queried cells that were rare.

Metrics and aggregation. Accuracy is the fraction of held-out cells classified correctly. Expected calibration error is the population-weighted gap between mean confidence and empirical accuracy over ten confidence bins. Rare-state recall is the fraction of true rare cells recovered (returning 1 by convention when a test set contains no rare cells, so a split is never spuriously penalised). All metrics are averaged over 5 seeds and reported with 95% confidence intervals (1.96 s.e.). All computation is CPU-only (NumPy, SciPy, scikit-learn, NetworkX, Matplotlib); a GPU build is a drop-in but not required.

Reproducibility. The full-scale run is launched with the command `scripts/run.py` on `configs/full.yaml` writing to `results/`; the file `summary.json` is the single source of truth for every table, figure and numerical claim, and Table 1 is transcribed verbatim from the generated `main_results.tex`. The provenance block records seed, platform, library versions, runtime (20.8 s) and peak memory (320.6 MB).

Data availability

No biological data were used or generated. All results derive from a seeded synthetic lineage that is regenerated deterministically by the code; the run artifacts (`summary.json` and `main_results.md` / `main_results.tex`) and figures are contained in the repository.

Code availability

The reference implementation (`topocell`), configurations, tests and scripts to reproduce every number, table and figure in this manuscript are provided in the repository accompanying this article. Honest ingest stubs for real AnnData and perturbation-screen data are included but are intentionally not exercised here.

Author contributions

M.H. designed the study, implemented the methods and experiments, analysed the results and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Supplementary Information

A from-scratch derivation of every concept the accompanying code implements. This appendix is written to be read top-to-bottom by someone who has seen linear algebra and basic probability but *not* single-cell biology, graph-based semi-supervised learning, the graph Laplacian, label propagation, calibration or active learning. Every symbol used in the reference implementation (`topocell`) is defined, and each subsection names the source file it maps to, so theory and implementation can be read side by side. The Methods section above gives the condensed version; here we expand it pedagogically without restating the experimental protocol.

S1 The problem in one paragraph

A single-cell experiment measures thousands of individual cells, each described by a vector of features (after dimensionality reduction, a low-dimensional embedding of its gene-expression profile). Cells do not fill this space uniformly: they lie near a low-dimensional curved surface (a “manifold”) that traces differentiation, activation or treatment response. The task we study is *forecasting*: predict the discrete state of a cell under a condition we did not directly observe—a donor not yet profiled, a later technical batch, a later point along the trajectory, or an untested perturbation. Two facts make this hard. (i) Only a handful of cells are ever annotated by hand; the rest are unlabelled. (ii) The biologically decisive states (transient progenitors, treatment-resistant clones) are often *rare*, and ordinary classifiers trained to minimise average error ignore them. The scientific question of this paper, kept deliberately narrow and falsifiable: does conditioning a forecaster on the *connectivity* of the cell k -nearest-neighbour (k NN) graph—rather than on point features alone—improve (a) generalisation to held-out groups and (b) recovery of a rare state? The answer is yes, but modestly on accuracy, more clearly on rare-state recall, and at a small calibration cost; all data are synthetic.

S2 The data: a seeded synthetic branching lineage

Source: `src/topocell/synthetic.py`. We do not ship or fabricate real biological data (see `ingest.py` for honest hooks into real `AnnData`). Instead we *generate* a fully controlled lineage with the same structure the method must handle, so every claim is checkable bit-for-bit.

A “lineage” is a tree of branches over a continuous coordinate $t \in [0, 1]$ called *pseudotime* (how far a cell has progressed). The tree here has three branches (`_TREE` in `synthetic.py`): branch 0 is the trunk/common progenitor, living over $t \in [0, 1]$; branch 1 is the first fate, which splits off the trunk at $t_0 = 0.45$; and branch 2 is the second fate, which also splits at $t_0 = 0.45$. Each branch has a fixed, well-separated *anchor* point in feature space. A cell on branch b at pseudotime t has a mean feature vector $\mu(b, t)$ that drifts linearly from the split point toward that branch’s anchor (function `_branch_mean`). The cell itself is

$$x = \mu(b, t) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \text{noise}^2 I).$$

The key geometry: early cells (small t) of different fates share the same trunk mean, so they *overlap* and are hard to separate, whereas late cells are well resolved. This is exactly the regime in which letting a label diffuse along the manifold should beat a point classifier.

Each cell also carries the metadata a real forecaster would use: **state** (the discrete label to predict, equal to the branch for the common branches); **donor** (a biological-replicate id); **batch** (a technical-replicate id); **perturbation** (a condition id, with 0 the control, applying a small additive feature shift that never erases the lineage); and **pseudotime** (the continuous progression coordinate). Donor, batch and perturbation are drawn *independently* of the biological branch, so the held-out splits of Section S7 are non-trivial (a donor spans multiple branches).

The rare state. A brand-new state label ($\text{id} = n_{\text{branches}}$) is carved out of the late tip ($t > 0.8$) of the second fate: a random `rare_fraction` of cells there are relabelled and additionally *nudged* into their own pocket of feature space ($X[\text{chosen}] += [1.5, 1.0, 0, \dots]$). With `rare_fraction` = 0.04 this is $\sim 4\%$ of cells—the transient subpopulation an analyst most wants to find. Everything is seeded with NumPy’s `default_rng(seed)`, so the lineage is bit-for-bit reproducible across machines.

S3 The graph object: the k NN cell graph

Source: `src/topocell/graph.py`. The bridge from “points in feature space” to “topology” is the k -nearest-neighbour graph, the same construction SCANPY uses before clustering or pseudotime [5]. To build it (`build_knn_graph`): for each cell, find its k nearest cells by Euclidean distance (self excluded), giving a *directed* k NN (an edge $i \rightarrow j$ if j is among i ’s k neighbours); then *symmetrise* with the either-direction convention $A \leftarrow \max(A, A^\top)$, so an undirected edge $\{i, j\}$ exists if i picked j or j picked i . Symmetrising is the standard single-cell convention and makes connected components well defined. The result is a symmetric adjacency matrix A ($A = A^\top$, $A_{ii} = 0$). A is the graph, and it is the *substrate* for everything that follows: labels diffuse over its edges, and rare cells are hunted in its sparse regions. The *degree* of cell i is $d_i = \sum_j A_{ij}$ (how many neighbours it has); collect the degrees on the diagonal of $D = \text{diag}(d_1, \dots, d_n)$.

S4 Topology: Betti-0 as a connectivity diagnostic

Source: `graph.py`. The only topological invariant the method uses is Betti-0, the number of connected components of the graph, equal to the rank of the 0-th homology group H_0 (`n_connected_components` computes it with SciPy’s `connected_components`). Intuitively, a connected component is a maximal set of cells reachable from one another by walking along edges; the number of them counts how many disjoint pieces the data splits into. A real, testable property: as k grows, edges are only *added*, so components can only *merge*—the component count is monotone non-increasing in k . `betti0_curve` returns this count for $k \in \{2, 4, 8, 16, 32\}$, and `tests/test_betti_monotonic.py` asserts the monotonicity as a build sanity check. Honesty note: at the reported scale the graph is a single connected component for every k in that set (the Betti-0 curve is constant at 1), so Betti-0 here is a *passive* sanity check, not an informative descriptor—which is precisely why the paper frames the method as graph-/manifold-structure-aware, not as deep topology (no persistent homology, no higher-order invariants). `topology_features` and `local_density` add cheap graph-level descriptors (mean degree, clustering, and a per-cell inverse-density proxy) used only by the active-sampling policy of Section S8.

S5 The graph Laplacian and Dirichlet energy

This is the heart of the mathematics. Define the (combinatorial) graph Laplacian $L = D - A$. Its single most important property is that, for any vector f on the cells,

$$f^\top L f = \frac{1}{2} \sum_{i,j} A_{ij} (f_i - f_j)^2 \geq 0.$$

This is the *Dirichlet energy*: it is small exactly when f varies smoothly across edges (neighbouring cells get similar values) and large when f jumps across edges. L is symmetric positive semidefinite; the all-ones vector lies in its null space (constant labels cost zero energy), and the multiplicity of the eigenvalue 0 equals the number of connected components (tying back

to Betti-0). “Smoothing labels over the graph” will mean choosing f that is smooth in this Dirichlet sense while still matching the few labels we have, made precise next.

S6 Label propagation as a minimum-Dirichlet-energy label extension

Source: `src/topocell/policy.py`. We observe one-hot labels Y on a small labelled set (mask M with $M_{ii} = 1$ if cell i is labelled, else 0) and want a soft-label matrix F (rows are cells, columns are classes). The variational objective—stated as Theorem 1 above—is

$$J(F) = \text{Tr}(F^\top L F) + \mu \|M(F - Y)\|_F^2.$$

The first term is the Dirichlet energy of Section S5 (smoothness on the graph); the second is a *fidelity* penalty (stay near the observed labels on labelled cells), with $\mu > 0$ the trade-off. J is convex; setting the gradient to zero gives

$$2LF + 2\mu M(F - Y) = 0 \implies (L + \mu M)F = \mu MY.$$

On any connected component containing at least one labelled cell, $L + \mu M$ is positive definite, so the minimiser is *unique*: label propagation is the unique minimum-Dirichlet-energy extension of the observed labels. The same theorem states the limitation—a component with no labels is not identified by smoothing alone—which is exactly the failure mode that active sampling (Section S8) attacks.

The actual iteration. The implementation runs the normalised form of Zhou et al. [1]. Symmetrically normalise the adjacency, $S = D^{-1/2}AD^{-1/2}$ (eigenvalues in $[-1, 1]$); seed F_0 as one-hot labels on labelled cells and zero elsewhere; and iterate

$$F \leftarrow \alpha(SF) + (1 - \alpha)F_0.$$

This is a damped power iteration whose fixed point $F^* = (1 - \alpha)(I - \alpha S)^{-1}F_0$ is the closed-form solution of a normalised version of $J(F)$, with α the diffusion/clamping trade-off. The code uses $\alpha = 0.9$ and 30 power iterations (deterministic, CPU, no learning). Held-out cells are predicted by the argmax of their row-normalised diffused scores; the maximum class probability is the *confidence* used for calibration (Section S9). This fixed-point solver is a faithful surrogate for a message-passing GNN: dropping in a learned propagator [14] leaves the rest of the pipeline unchanged.

The honest baseline. The `KNNForecaster` is a plain k -nearest-neighbour classifier in feature space, fitted on the *same* labelled cells with the *same* k , but with *no* diffusion. It sees identical neighbourhood information; the only difference is that it does not let labels spread along the graph. Comparing `LabelPropagation` to this baseline isolates the value of graph diffusion, not the mere availability of neighbours—the apples-to-apples control that is the methodological spine of the paper.

S7 Splits, leakage and the few-label regime

Source: `src/topocell/splits.py`. A forecaster that only works when train and test share donors, batches, conditions or timepoints is a *memoriser*, not a forecaster. So we evaluate transfer across nuisance variables under four protocols (`SPLIT_TYPES`): `donor` (hold out whole donors—biological-replicate transfer); `batch` (hold out whole batches—technical-replicate transfer); `time` (train on early pseudotime $t < \text{cutoff}$, test on late—a genuine forward-in-time extrapolation); and `perturbation` (hold out a whole condition—forecast an unseen perturbation).

For grouping splits, `has_leakage()` checks that the held-out group’s id appears in test but not in train (disjoint group sets), and the test suite asserts this strict family-held-out no-leakage property. The time split cuts on a continuous coordinate, so there is no grouping variable to leak (clean by construction).

Few-label annotation. Only a fraction `label_fraction` of the train pool is actually annotated (`_annotate`), mirroring how analysts label a handful of cells and propagate the rest. The selection is lightly *stratified*: a couple of labels are guaranteed for each present state, including the rare one—with zero rare labels no method could recall the rare state and the comparison would be uninformative rather than honest. The held-out test cells are never annotated.

S8 Active sampling: hunting the rare state

Sources: `policy.py`, `graph.py`. Active learning chooses *which* cells to annotate next, given a fixed labelling budget. The topology-aware policy (`active_sample`, `strategy='topology'`) scores each unlabelled cell by

$$\text{score} = \text{zscore}(\text{uncertainty}) + \text{zscore}(\text{inverse local density}),$$

and greedily takes the top `budget` cells. The two ingredients are: (i) *uncertainty*, the Shannon entropy $H(p) = -\sum_c p_c \log p_c$ of a label-propagation forecast from the *current* labels (`_entropy`), where high entropy means the model is unsure—near a decision boundary, the classic active-learning signal; and (ii) *inverse local density*, the reciprocal of a cell’s mean *k*NN distance (`graph.local_density`), since rare states sit in low-density pockets, making $1/\text{density}$ a cheap, topology-aware “where are the rare cells” signal. Standardising each to a z-score (subtract mean, divide by standard deviation) puts the two heterogeneous signals on a common scale before adding them. The baseline is uniform *random* querying (`strategy='random'`). `recall_after_query` measures rare-state recall on the still-unlabelled cells before versus after adding the queried set; the gap is the active-sampling *lift*. We also report the fraction of queried cells that were genuinely rare (`rare_query_rate`).

S9 Metrics and aggregation

Source: `src/topocell/metrics.py`. *Accuracy* is the fraction of held-out cells classified correctly. *Expected calibration error* (ECE) asks whether the confidences are trustworthy: bin the test cells by predicted confidence into ten bins, in each bin compare mean confidence to empirical accuracy, and take the population-weighted average gap; a perfectly calibrated forecaster has $\text{ECE} \rightarrow 0$. We report ECE precisely because graph smoothing sharpens probabilities and, in this setting, makes them mildly overconfident—a small, honestly disclosed cost that recalibration in the manner of Guo et al. [16] would remove. *Rare-state recall* is the fraction of true rare cells the model labels rare; by convention it returns 1 when a test set has no rare cells (nothing to miss), so a split is never spuriously penalised. *Aggregation* (`summarize`): every metric is computed per seed and reported as mean, sample standard deviation s (with $\text{ddof} = 1$) and a 95% confidence interval $1.96 s / \sqrt{n}$ over the $n = 5$ seeds. These `ci95` values are exactly the error bars in the figures and the $\pm$ in the text.

S10 Reproducibility object: one source of truth

Sources: `runner.py`, `scripts/`. `runner.py` ties everything together: generate the lineage, build the graph, check integrity (leakage-clean splits, rare state present, Betti-0 monotone), run both

forecasters under all four splits over five seeds, run the active-sampling comparison, and write `results/summary.json`—the single source of truth. Nothing in the paper is typed by hand:

```
summary.json  $\xrightarrow{\text{make\_tables.py}}$  results/main_results.tex,
summary.json  $\xrightarrow{\text{make\_figures.py}}$  figures/fig_{schematic,lineage,forecast}.pdf.
```

`scripts/audit_claims.py` is a readiness gate: it checks that every headline number is present and traceable, that the integrity flags are true, and (against the repository README) that the mandated reproducibility phrase is present with no forbidden superiority claims; it exits non-zero if any check fails. Figures are drawn by `src/topocell/plotting.py` in *Nature Machine Intelligence* house style (vector PDF with embedded fonts, sans-serif, colour-blind-safe Okabe–Ito categories with viridis for the continuous embedding axis, bold lower-case panel labels, 95% CI error bars, no in-plot titles); `fig_schematic` is the method-overview Figure 1.

To reproduce everything from the `code/` folder:

```
export KMP_DUPLICATE_LIB_OK=TRUE
pip install -e . # or: pip install -e .[gpu]
bash scripts/reproduce_all.sh full # writes summary.json, tables, figures
```

The reported run uses 3000 cells, 16 features, 8 donors, 4 batches, 5 perturbations, 3 branches, ~4% rare state, $k = 20$, `label_fraction = 0.03`, `active_budget = 60`, and 5 seeds with master seed 0; the provenance block records a runtime of 20.8s and peak memory of 320.6 MB on a laptop CPU. The figure and table artifacts are copied verbatim into the `submission/` folder used to build this manuscript.

S11 What the result means (and what it does not)

Within this controlled synthetic benchmark, conditioning forecasts on k NN-graph connectivity helps: label propagation beats the matched point classifier on held-out accuracy on all four splits (mean gain ≈ 0.0595 , roughly four to nine percentage points) and—the clearer effect—recovers a rare state the baseline almost never finds. The topology-aware active-sampling policy raises rare-state recall from 0.0247 (random) to 0.3556. These gains carry a small, consistent calibration penalty (ECE is slightly worse on every split). The accuracy gain is real but modest; the headline contribution is rare-state recovery, and even that is null on the time split (forward-in-time extrapolation puts the rare late-tip cells entirely outside the training distribution—reported as-is). The critical caveats, expanded in the Limitations above, are that all data are synthetic (no biological discovery claim); that the “topology” is shallow (k NN-graph connectivity and a Betti-0 component count that is constant at this scale, not persistent homology); that rare-metric confidence intervals are wide because rare cells are few; and that a single synthetic generator cannot substitute for real datasets. The decisive next step is to repeat this exact comparison on real scRNA-seq and perturbation-screen data via the unexercised ingest stubs.